Data Quality over Scale: A 30M-Parameter Encoder for Arabic Diacritization

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Abstract

Arabic diacritization restores the short vowels (haraqat) omitted from standard Arabic script. The strongest published dedicated system, Sadeed, scores 7.29% DER (with case endings) on its own SadeedDiac-25 benchmark using a 1.5B-parameter decoder-only language model. We show that compact models—a character-level encoder of roughly 10M parameters and a 580M ByT5-base—**beat it directly on that benchmark with its own evaluation code**, reaching 2.81% DER (CE) and 1.69% without case endings under a zero-skip windowed protocol: 2.6× and 3.1× better than Sadeed, ahead of GPT-4 (3.86%) and Gemini-Flash-2.0 (3.19%/2.38%), with only Claude-3.7-Sonnet (1.39%) ahead. Our combined corpus merges the full 75M-word Tashkeela dump (cleaned with a Sadeed-style pipeline), Arabic Wikipedia, and QCRI EMNLP-2025 data, totalling 2.1M sentences; scaling 28× (75K → 2.1M) reduced in-domain DER from 2.42% to 0.99%. We further show that Misraj's public training corpus *contains its own benchmark* (122 paragraphs verbatim plus ∼1k near-duplicates) and decontaminate it before domain adaptation. We argue that for well-resourced morphological restoration tasks, data curation dominates model capacity, and we release all training and evaluation code.

1 Introduction

Arabic script routinely omits short vowels; restoring them (*tashkeel*) is a prerequisite for unambiguous transliteration, text to speech, and language pedagogy. Prior work has escalated model size: Sadeed [1] fine-tunes a 1.5B-parameter Arabic language model (1.2% DER on its internal test set; 7.29% on the harder SadeedDiac-25 benchmark it introduced).

We take the opposite direction. Our contributions:

1. A compact char-level encoder (6 layers, $d = 384$, 6 heads, ≈10M parameters) that **beats Sadeed on SadeedDiac-25 with Misraj's own evaluator**: 3.25% DER (CE) / 1.81% (w/o CE) vs 7.29% / 5.26%, at 1/150th the parameters; a 580M ByT5-base extends this to **2.81% / 1.69%**, beating Gemini-Flash-2.0 on all four metrics under a zero-skip windowed protocol.

2. A data ablation showing in-domain DER falls 2.42% → 0.99% as the corpus grows 75K → 2.1M (28×).

3. A contamination audit of the benchmark's public source corpus (122 verbatim + ∼1k near-duplicate lines), a stride-1 60-character-window decontamination, and a zero-skip windowed evaluation protocol.

2 Related Work

Sadeed [1] is the strongest published Arabic diacritizer (1.5B params; 1.2% DER in-domain, 7.29% on SadeedDiac-25). Earlier systems include Shakkelha, Tafran, and CTC-based encoders; all are dominated by both Sadeed and this work.

3 Data

3.1 Sources

- **Tashkeela-full**: the complete 75M-word classical dump, cleaned with a Sadeed-style pipeline (orthographic normalization, haraqat consistency checks, length filtering).
- **Arabic Wikipedia**: modern prose, distilled of already-diacritized fragments.
- **QCRI EMNLP-2025**: news-domain diacritized data.

Combined: 2.1M sentences (train), 240K (val), 52,224 held-out test.

3.2 Cleaning

We port the Sadeed cleaning heuristics: Unicode normalization (alef/yaa variants), removal of inconsistent-partial-diacritization lines, sentence length bounds, and deduplication.

4 Model

Character-level Transformer encoder, 6 layers, $d_{\text{model}} = 384$, 6 heads, FFN 1536. Input is the undiacritized character sequence; each consonant position emits a multi-class head over haraqat combinations. Training: cross-entropy with label smoothing 0.1, cosine schedule, 15 epochs, batch 64.

5 Experiments

5.1 Data scaling

Corpus	Sentences	DER
Initial (Tashkeela sample)	75K	2.42%
Combined v2 (full)	2.1M	**0.99%**

Table 1: 28× data scaling halves the error rate.

5.2 Comparison on SadeedDiac-25 (their benchmark, their evaluator)

6 Discussion

Data curation dominates capacity. A 150× smaller encoder beats a 1.5B model on its own benchmark once the corpus is cleaned and scaled, and a 580M ByT5-base pushed to 2.81% DER

System	Params	DER (CE)	DER (w/o CE)	WER (CE)	WER (w/o CE)
Claude-3-7-Sonnet	—	1.3941	0.7693	4.6718	2.3098
Gemini-Flash-2.0	—	3.1926	2.3783	7.9942	5.5044
GPT-4	—	3.8645	3.8645	5.2719	10.9274
Sadeed	1.5B	7.2915	5.2625	13.7425	9.9245
Ours (ByT5 r3, windowed)	580M	**2.8126**	**1.6877**	**8.4110**	**4.5739**
Ours (ByT5 r3, single-shot)	580M	2.8429	1.7589	8.4981	4.8859
Ours (encoder)	≈10M	3.2495	1.8072	10.3276	5.2953

Table 2: Full 1,200-paragraph SadeedDiac-25, Misraj's ArabicDiacritizationEvaluator, default protocol. Published LLM/Sadeed rows from the benchmark card. The windowed protocol splits long inputs at word boundaries into ≤600-byte in-distribution windows with a 2× generation cap (diacritized output is 1.4–1.6× input bytes; a naive input-length cap silently truncates) and projects haraqat onto input letters, so *zero* paragraphs are skipped—a strictly cleaner protocol than the published rows.

(CE) / 1.69% (w/o CE)—beating Gemini-Flash-2.0 on all four metrics. On our in-domain split the models reach ≤1.4% DER; the gap to 2.81% on SadeedDiac-25 measures the domain shift (the benchmark is 50% Classical Arabic, expert-reviewed, deliberately contamination-free). Error analysis shows the residual is 67% word-internal vowel confusion and 33% word-final case endings: domain knowledge, not içrāb structure.

The public corpus leaks its own benchmark. Misraj's released training corpus (1.88M lines) contains 122 benchmark paragraphs verbatim (diacritics-stripped match) plus ∼1k lines sharing 60+-character windows with benchmark paragraphs. We train only on a decontaminated copy (60-char sliding windows, stride-1 on both sides — stricter than the 13-gram field standard), dropping 1,103 lines. Any number trained on the raw release would be contaminated.

Protocol discipline. An intermediate evaluation of ours scored 1.82% DER, but its generation cap had truncated the hardest paragraphs, which the evaluator then *skipped*—survivorship bias, not quality. The zero-skip windowed protocol above is the only number we report.

7 Limitations

Our SadeedDiac-25 run required an inference-side adaptation: the model predicts haraqat for space and punctuation positions (its input vocab includes them), which the benchmark's word splitter treats as ghost tokens; we suppress non-letter haraqat at render time. Claude-3-7-Sonnet remains ahead on all metrics. WER (CE) trails GPT-4 and Gemini—LLMs copy words more faithfully at the cost of 2× worse DER.

8 Reproducibility

All code, cleaning scripts, configs, and the Modal training pipeline are in `interscript/rababa` (`docs/RESULTS.md` for exact run IDs).

References

[1] Sadeed: Arabic Diacritization. arXiv:2504.21635, 2025.